



A DEEP DIVE INTO SUPPLEMENTS ADVERSE EVENTS

By Ali Albinfalah

Introduction

The FDA monitors food and dietary supplement safety through the **Human Foods Complaint System (HFCS)**, formerly known as **CAERS**. The system is used to **track adverse event reports** and identify long term safety trends across food and dietary supplement products.



Problem Statement

Between **1999 and 2025**, the FDA recorded over **90,000** adverse event reports linked to food and dietary supplements. Unlike pharmaceuticals, supplements **do not require FDA approval** before reaching consumers making post market reporting the only safety net.

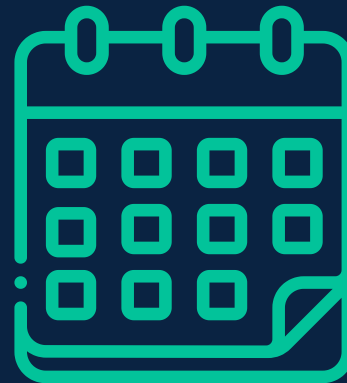


Dataset Overview



Data Source

FDA Human Foods
Complaint System (HFCS)



Time Period

1999 - 2025



Total Reports

96k

Objectives

1

Change in adverse
event reports from
1999–2025

2

Product categories
with the highest
reports

3

Influence of gender
and age on
supplement reporting

4

Most commonly
reported symptoms
among supplement
reports

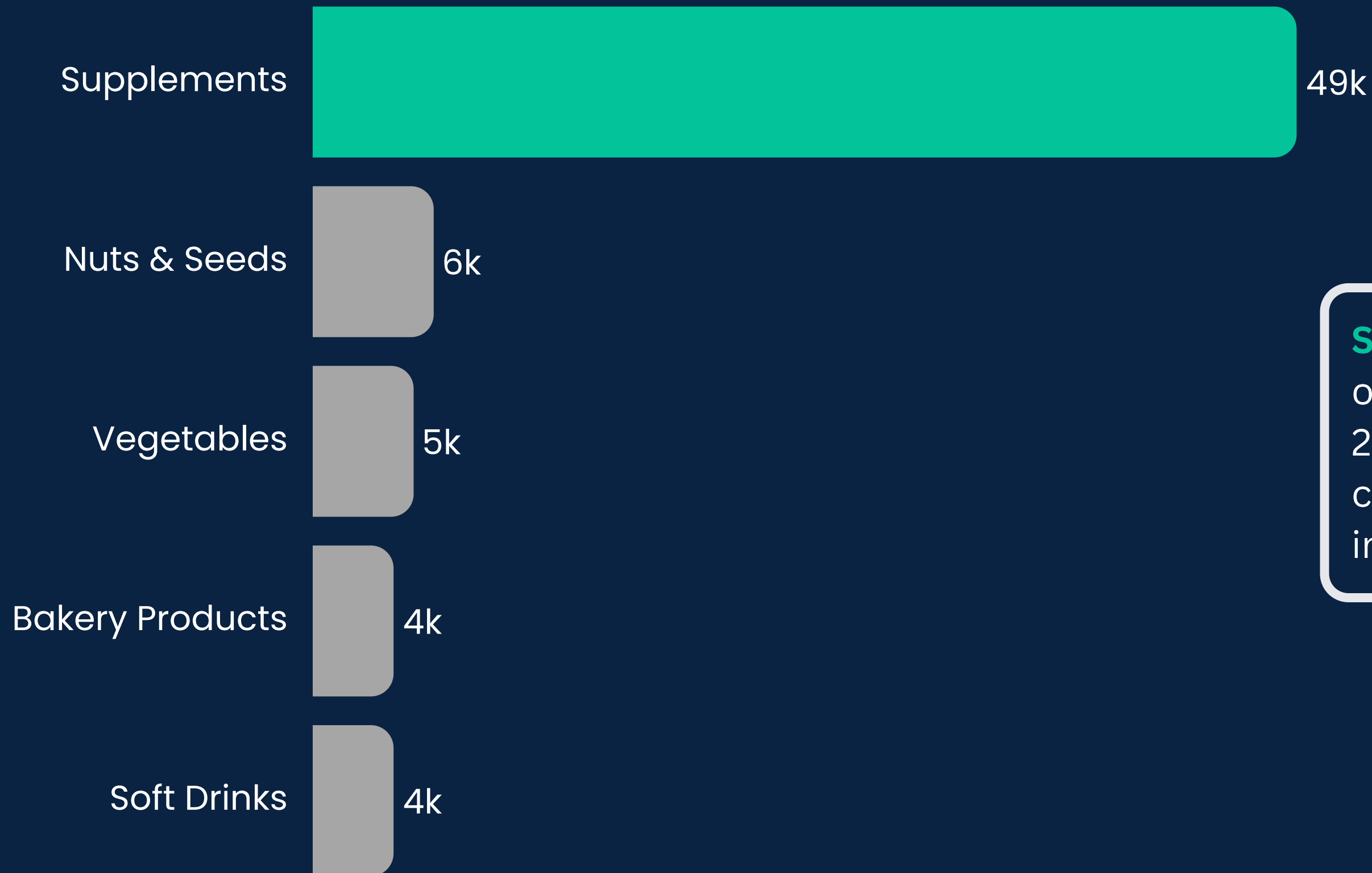
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Supplements most
frequently linked to
death reports

Reports Surged After 2007 Mandatory Reporting Law

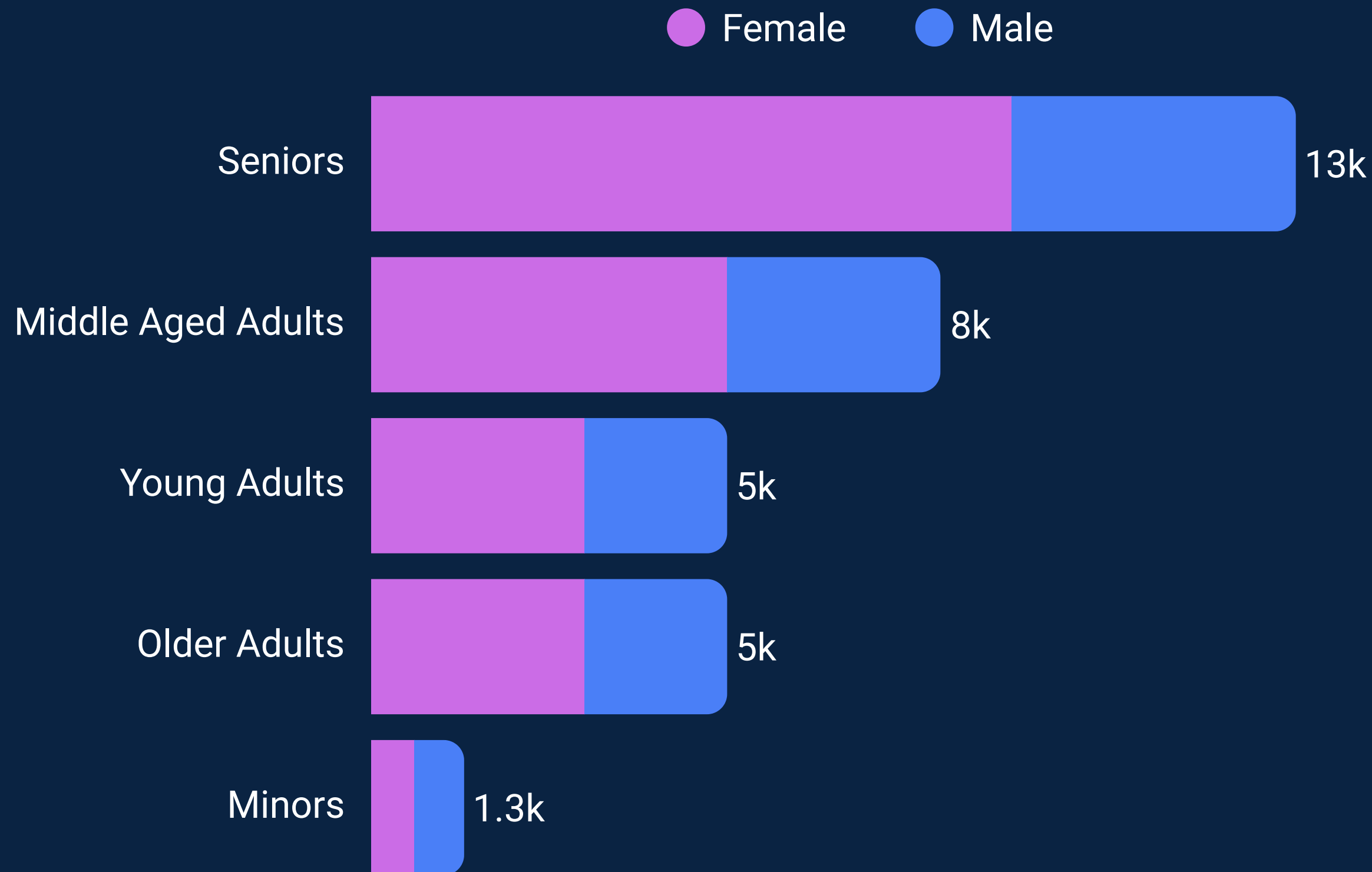


Supplements Dominate Reports by a Wide Margin



Supplements drive the vast majority of reports. This is primarily due to the 2007 mandatory reporting law and complex drug - supplement interactions.

Females and Seniors Are the Most Affected – Supplements

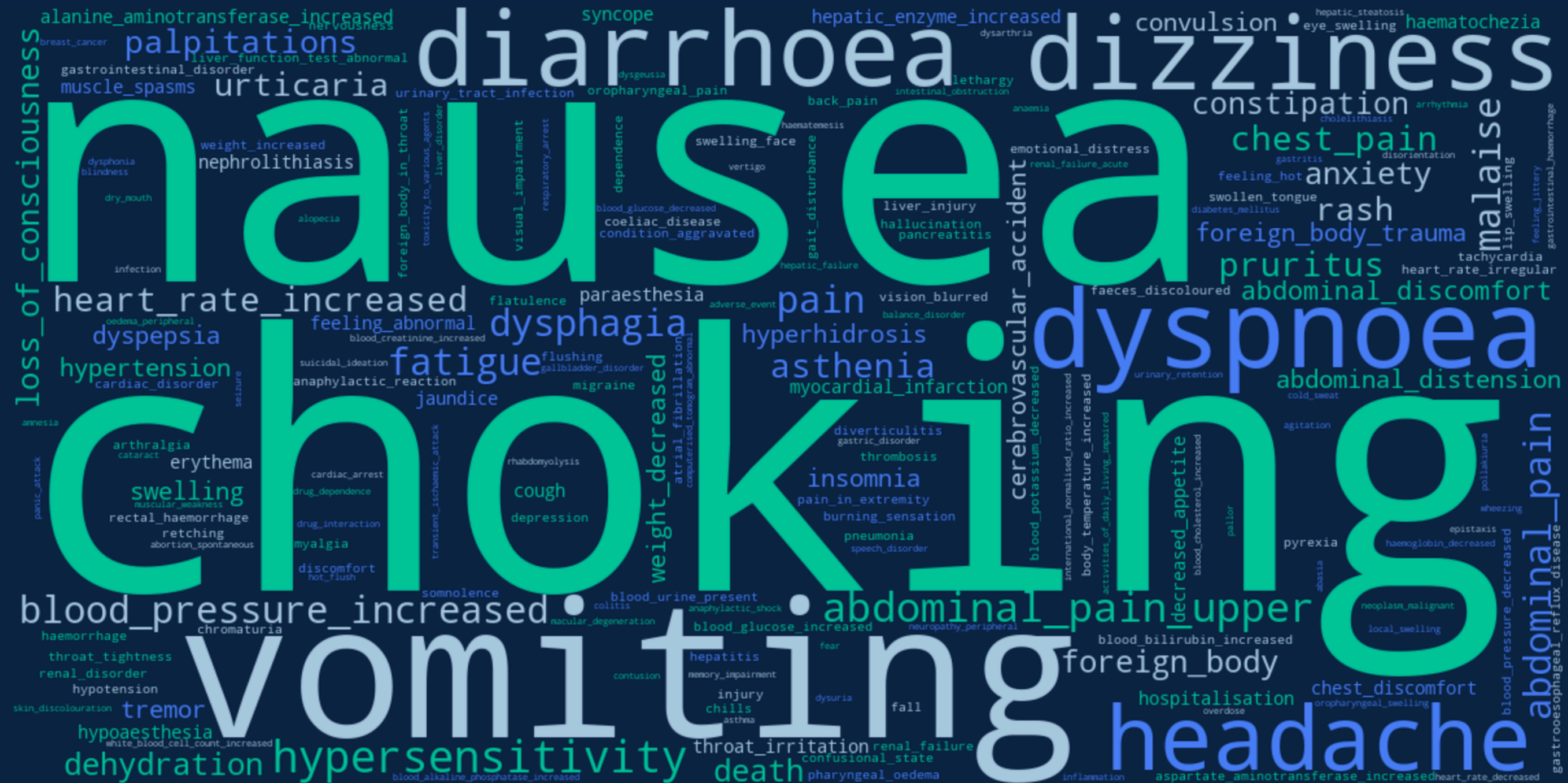


Females & Seniors (65+) are the primary reporters.

- Gender Gap: Females outnumber males in every age group.
- Senior Risk: High volume due to medication interactions.

Note: Gender was not reported in ~9% of cases and age in ~40% of cases

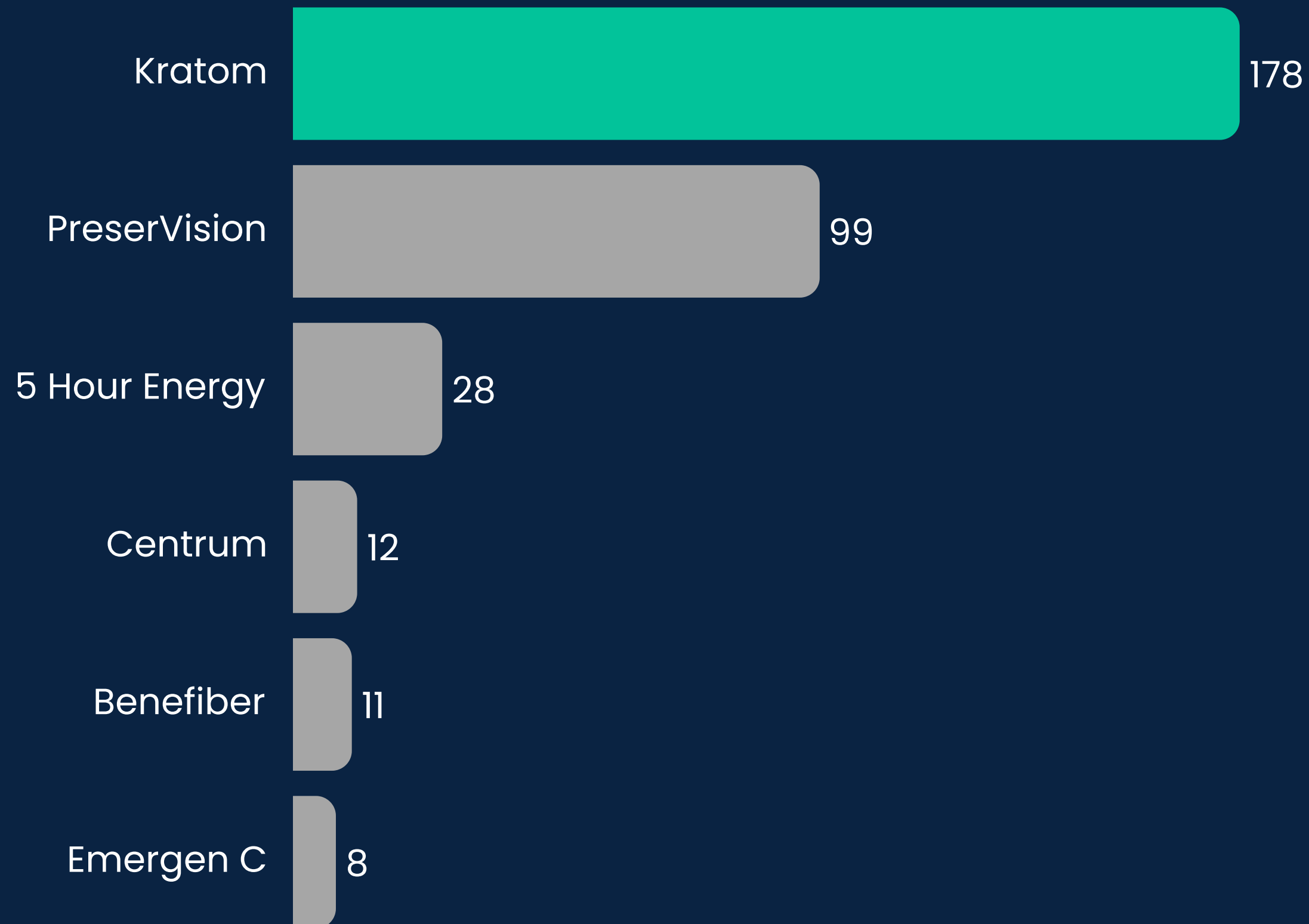
Most Reported Symptoms – Supplements



Symptoms & Their Associated Supplements

Symptom	Associated Supplements
Choking	Large multivitamins, fish oil softgels, calcium tablets
Nausea / Vomiting	Iron, Zinc (on empty stomach), high dose Vitamin C
Diarrhea	Magnesium, high dose Vitamin C, fiber supplements
Dizziness	Preworkouts, weight loss supplements (Hydroxycut)
Headache	High dose Vitamin D, weight loss supplements (Hydroxycut), preworkouts

Kratom Is Associated with the Most Reported Deaths



- **Kratom** - opioid like effects, multiple FDA safety warnings issued.
- PreserVision - correlated with the elderly demographic.
- 5-Hour Energy - high caffeine content linked to cardiovascular events.

SUMMARY



Supplements
account for 51%
of all adverse
event reports



Females and
Seniors (65+) are
the most affected
demographic



Choking and
gastrointestinal
symptoms are the
most reported
adverse effects



Kratom is
associated with
the most reported
deaths

RECOMMENDATIONS

Screen patients
for herbal
supplement use

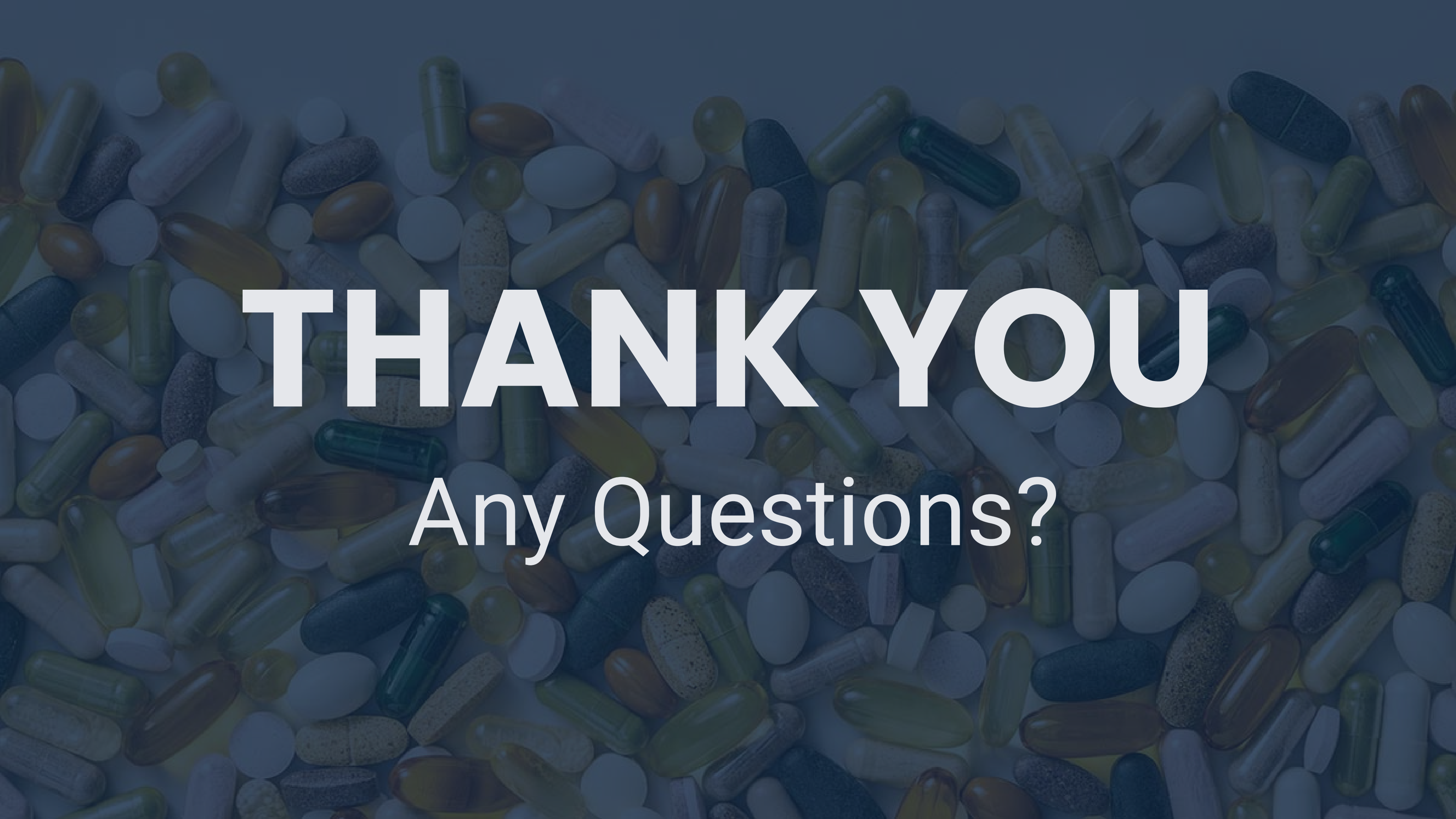
Prioritize
supplement safety
discussions with
seniors

Address choking
risk with elderly
patients

Use HFCS as a
clinical reference

LIMITATIONS

- Reports are self reported and not confirmed proof that the product caused the event.
- Side effects may be linked to other health conditions.
- Report quality varies, some reports are incomplete or missing key details.
- Incomplete contact information limits FDA's ability to fully investigate reported events.

The background of the image is a dense, overlapping collection of various pills and capsules. The pills are in many different shapes, including round, oval, and capsule forms, and come in a wide range of colors such as white, yellow, green, brown, and blue. They are scattered across the entire frame, creating a textured, medical-themed background.

THANK YOU

Any Questions?